

SIMULATION OF HYDRATED DMPC PHOSPHOLIPID MEMBRANE

EXPLORING THE DYNAMICS: ROTATIONAL CORRELATION OF WATER DIPOLES
DEPENDING ON DISTANCE AND TEMPERATURE

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Abstract

We present a molecular dynamics (MD) simulation study focused on the dynamics of water molecules in a hydrated DMPC phospholipid membrane. Using Langevin dynamics and a specialized force field, we investigate the rotational correlation of water dipoles as a function of temperature and distance from the membrane surface. Our results reveal distance-dependent dipole rotational autocorrelation, with increased correlation near the membrane interface. Additionally, temperature impacts on the system with thermal fluctuations, with core rotational correlated dynamics at lower temperatures. These findings deepen our understanding of water-lipid interactions and have implications for membrane-related processes and biomaterial design.

I INTRODUCTION

The present project focuses on investigating the behavior of a lipid bilayer solvated by water molecules. Understanding the dynamics and interactions between lipids and water is essential for comprehending the complex properties of biological membranes. The goal is to reach some conclusions from analyzing the dipolar rotational autocorrelation function of water molecules in the neighborhood of the lipid bilayer, considering variations in distance, what can be interpreted as solvation shells, and temperature.

Lipid bilayers, as a kind of micelles, are fundamental components of cell membranes and play a crucial role in maintaining the integrity and functionality of cells. It acts as a barrier to prevent the contents of a cell from dispersing and also to exclude external agents such as viruses. The membrane, however, does not have a purely passive role. It also enables the transport of ions and chemicals such as proteins, sugars and nucleic acids into and out of the cell via the membrane proteins. The lipid bilayer's unique structure and properties stem from the arrangement of phospholipids, which consist of hydrophilic heads and hydrophobic tails. These phospholipids hydrophilic heads interact with water and hydrophobic tails form the

core of the lipid bilayer [1]. Therefore, there is an antiparallel orientation of lipid layers in the bilayer that is maintained due to the extremely slow ‘flip-flop’ rate [2]. Water molecules surround the lipid heads, forming hydrogen bonds with them, which contribute to the overall structure and properties of the membrane [3].

The dipolar rotational autocorrelation function is a powerful tool for examining the rotational dynamics of water molecules in their local environment. By measuring the correlation between the dipole orientations of water molecules at different time intervals [4], we can gain insights into their mobility, orientational ordering, and interactions with the lipid bilayer.

In this project, we will employ molecular dynamics simulations to simulate a hydrated DMPC phospholipid membrane system. By systematically analysing different distance ranges between the lipid bilayer and water solvation, as well as exploring different temperatures, we can investigate the effects of these factors on the dipolar rotational autocorrelation function of water molecules. Based on the understanding that the lipid heads form hydrogen bonds with water, we expect to observe a stronger autocorrelation near the lipid heads compared to regions further away. This localized autocorrelation is indicative of the enhanced interactions between water molecules and the lipid head

groups.

The findings of this project will contribute to our knowledge of the interplay between lipids and water in biological systems. Furthermore, they may have implications in various fields such as drug delivery systems, membrane biophysics, biomaterial design, and cosmetics [2]. By elucidating the dipolar rotational autocorrelation function in the solvation shells surrounding the lipid bilayer, we can advance our understanding of the complex dynamics of membranes and pave the way for the development of novel biomaterials with tailored properties.

II DESCRIPTION OF THE SYSTEM AND OBSERVABLES

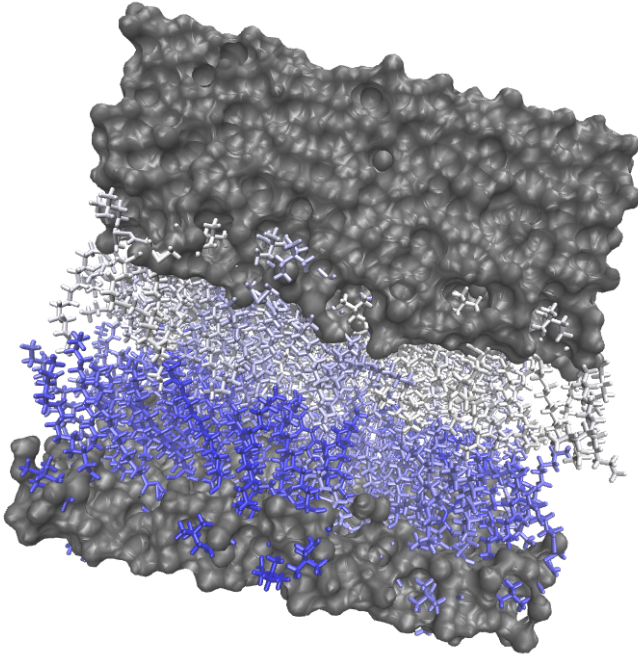


FIGURE II.1: Visualization of the system using VMD software. Waters are represented in grey and lipids in blue and white.

As it has been briefly explained in the introduction a lipid bilayer is formed by a two lipid monolayers with antiparallel orientation. That way the hydrophobic tails remain in the inner part of the layer without being hydrated. This formation is possible due to the self-assembly generated by the hydrophobic effect, a phenomenon that has been thoroughly studied for this system in the literature [1, 2]. The DMPC lipids that conform the bilayer are phospholipids incorporating a choline as a headgroup and a tailgroup formed by two myristoyl chains. Choline based phospholipids are ubiquitous in cell membranes and used in drug targeting liposomes. From this lipid, the phosphate group is H-bond acceptor so H-bonds are probably formed

between that group and the water molecules. A visualization of the system and a schematic representation are provided in Fig. II.1 and Fig. II.2. In that representation the water molecules are described as the grey surface and there are surrounding both sides of the lipid bilayer. The lipids are represented as chains and each of the monolayers is colored with different color, blue or white.

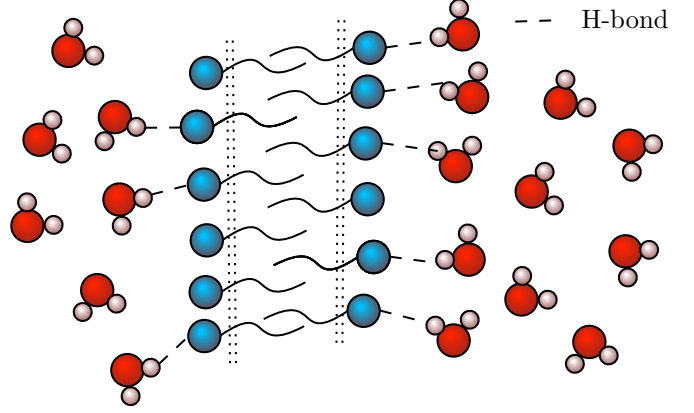


FIGURE II.2: Diagram representation of the membrane-water system

The observable utilised to analyse the system will be the dipolar rotational autocorrelation, that calculates the difference in the dipole vector direction $\hat{\mu}$ at different time steps. As it is autocorrelation this observable reflects the change in the orientation of the water molecule as the time advances. The rotational dipolar autocorrelation function is defined by

$$C_{sim}^{rot}(t) = \langle \hat{\mu}(t) \cdot \hat{\mu}(0) \rangle \quad (\text{II.1})$$

where $\hat{\mu}$ is the dipole vector direction and $\langle \dots \rangle$ denotes average value over all particles. The water dipole vector $\hat{\mu}(t)$ is given by the direction which bisects the angle between the directions of the oxygen atom and the two hydrogen atoms of a water molecule.

As part of the project is to analyse this dipolar rotation autocorrelation for molecules at different distance range of the membrane, the water molecules have been filtered by VMD software by introducing `water within ? of the lipid` where the `?` is a notation to indicate any digit that represents a distance in angstroms. As this method does not select water molecules but its atoms, some molecules are truncated in the border. To save this shortcoming VMD has been used again and it has been used a select atoms by the keywords `within 1 of numbonds=2` that will filtrate only the complete water molecule in the range of interest. Some molecules, those that are truncated, do not count to the calculation, but the fraction is so little that does not affect the results.

III TEMPERATURE DEPENDENCE ON THE DIPOLAR ROTATIONAL AUTOCORRELATION

The dipolar autocorrelation function will be calculated for different temperatures in a range from $T = 290$ K to $T = 310$ K. By statistical physics arguments it is expected that for higher temperatures there will be more thermal fluctuations and the autocorrelation will be then lower, since the water molecules will have more random orientation as time passes.

The dipolar rotational autocorrelation for all water molecules in the system and different temperatures can be found at Fig. III.1. In the light of the results, the autocorrelation at large values of time is, in fact, lower when a higher temperature is computed, specially for the highest temperatures $T = 300$ K and $T = 310$ K. For the temperatures below $T = 300$ K the autocorrelation seems to reach similar values for large times.

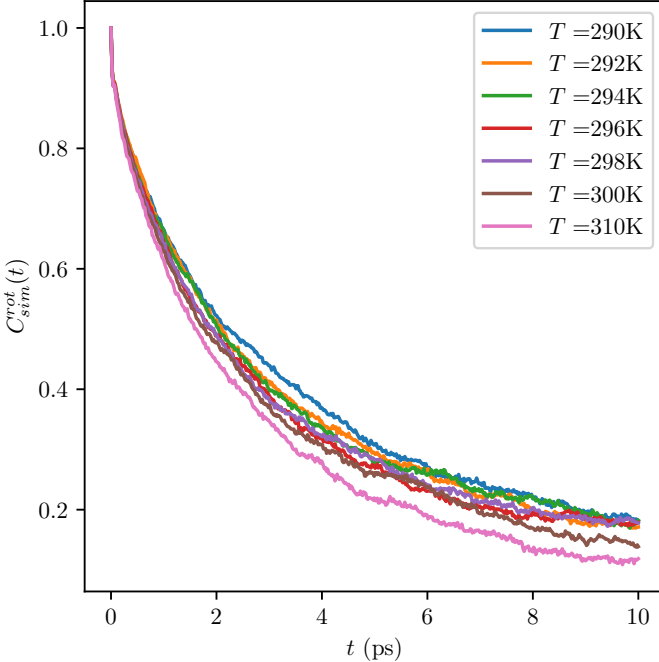


FIGURE III.1: Dipolar rotational autocorrelation for different temperatures for all molecules.

IV ORIENTATION OF WATER DIPOLES FOR DIFFERENT DISTANCE RANGES AND TEMPERATURES.

Using all the molecules in the calculation will give an average insight of the water orientation of the system, but, since there is an interaction between the hydrophilic heads and the hydrogen of the waters, it is expected that the autocorrelation will be higher near the bilayer and lower when the interaction lipid-water is no longer important. Therefore, computing the autocorrelation for some intervals of the distance will give valuable information of the system. In Fig. IV.1 is shown the curve of the autocorrelation for different range distance and temperatures $T = 290, 294, 296, 300$ and 310 K. These temperatures have been chosen because they are meaningful in the range analysed, in Fig. III.1 can be found that temperatures $T = 292$ K and 298 K have similar behavior to $T = 294$ K and 296 K. For the nearest molecules (distance lower than $3 \text{ \AA}$) the autocorrelation at long time remains between 0.4 and 0.6 approximately, while for the further distance range (distance greater than $9 \text{ \AA}$) the autocorrelation in the long term reaches zero. Therefore, this is an indication that the membrane has an interaction with the water molecules that impedes the water dipoles from reorienting too much. Furthermore, the autocorrelation of the intermediate range (distance greater than $3 \text{ \AA}$ but lower than $9 \text{ \AA}$) reassures that the interaction is weaker when further away from the membrane and, also, falls rapidly since the autocorrelation of the intermediate range is more similar to the furthest range than the nearest. This is consistent with an interaction of hydrogen bonds and pair charge, since they are interactions of molecular scale. Also, it is proved that using all the molecules is tantamount to an average measure.

In the Fig. IV.1, the similarities of behaviour for different temperatures are appreciated, but there is no information about how the temperature affects the autocorrelation of each distance range. To analyse that the autocorrelation for different distance range and temperatures is plotted altogether.

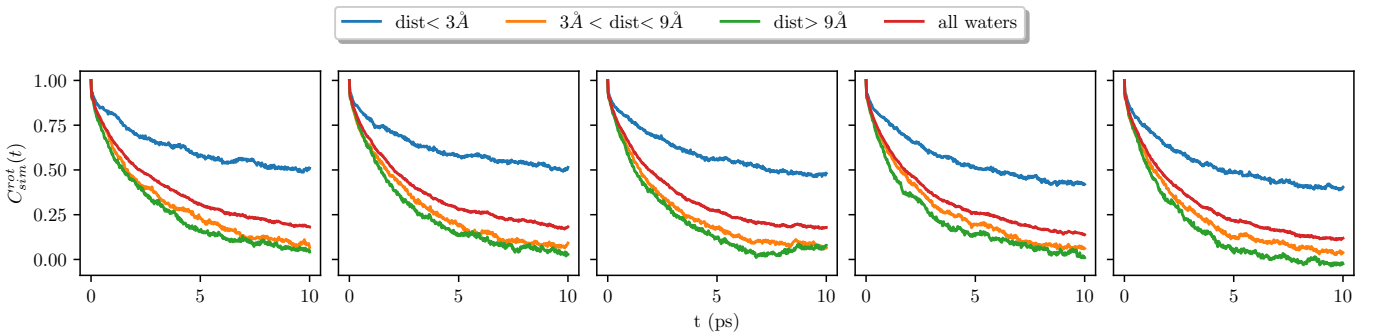


FIGURE IV.1: Comparison for the autocorrelation curve for the different range distance and with all the molecules. From left to right $T = 290, 294, 296, 300$ and 310 K.

In the Fig. IV.1, the similarities of behaviour for different temperatures is appreciated, but there is no information about how the temperature affects the autocorrelation of each distance range. To analyse that the autocorrelation for different distance range and temperatures is plotted altogether. In Fig. IV.2 the autocorrelation for the nearest and the intermediate range has been depicted for different temperatures. In the first case, the autocorrelation for the lower temperatures 290 K and 294 K is almost the same, but for greater temperatures the difference is notorious, specially for $T = 296$ K. This difference in behaviour in a two K distance could be related with a limit temperature for what the kinetic energy of the water can compete with the interaction of the hydrogen bond, although the data are not enough to make possible a conclusive statement. On the other hand, in the intermediate range, the values of the autocorrelation at large time is similar. That is probably because all the value are almost zero in the long term, and the behaviour is almost completely dominated by the thermal fluctuations. As the change in temperature is not high, there is 20 K difference at most, the thermal fluctuation does not produce significant change.

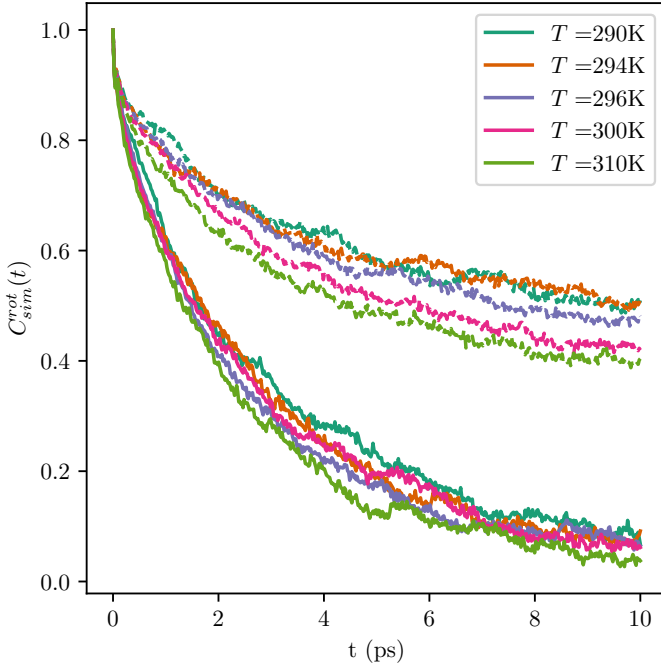


FIGURE IV.2: Dipolar rotational autocorrelation for different temperatures for the nearest range distance lower than $3 \text{ \AA}$ (dashed lines) and the intermediate range, distance greater than $3 \text{ \AA}$ but lower than $9 \text{ \AA}$ (solid line). The furthest range is not plotted because it give similar information that the obtained with the intermediate range.

V EFFECT OF A SIMULATION REPETITION

For the temperature $T = 300$ K the simulation has been repeated twice, doing a different initial stabilization and simulation (see section VI for a deeper explanation about the parts and details of the simulation). Since the molecular simulation has been done using Langevin dynamics, the trajectory of the system relies in stochastic process and; therefore, the result of the stabilization is different in each realization. Since a repetition for the same temperature has been done, but it has not been used the same initial condition nor the seed of the semialeatory numbers generator, we can explore a broader range of possibilities and gain a more comprehensive understanding of the system's behavior. This approach helps to mitigate the influence of specific initial conditions on the results and provides a statistical representation of the system's properties. Also, it smooth the curves of the magnitudes computed, since due to various factors such as inherent uncertainties in initial conditions and the influence of thermal fluctuations, the trajectories of individual particles can exhibit significant variability, leading to noisy or erratic behavior in the observed magnitudes. For example, in section IV the noise of the curve impede to have data reliable enough to try finding a norm or tendency in the variation of the temperature for the autocorrelation in the nearest range. A way to obtain reliable data will be doing several repetitions and reduce the noise by averaging.

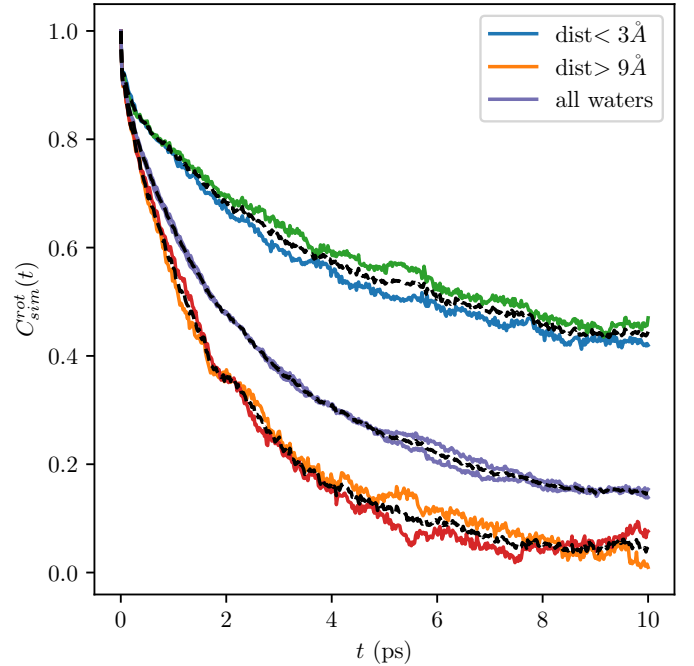


FIGURE V.1: Dipolar rotational autocorrelation for the temperature 300 K repeated twice with different initial conditions. Black dashed lines represent the average values of the different curves.

To see this effect, the Fig. V.1 has been created, using two repetitions of the simulation at $T = 300$ K. As it can be seen both repetitions does not give the same results, although it is indeed similar. The black dashed lines has been plotted to represent the average value of both curves. Although they are only two repetition the black lines are obviously smoother, for a good result averaging over some tens of repetitions will be required.

In conclusion, the repetition of molecular dynamics simulations with different initial conditions and the use of Langevin dynamics is needed for smoothing the curve of magnitudes. This approach takes into account the stochastic nature of molecular systems, incorporates thermal effects, and provides a statistical understanding of the system’s properties. By conducting multiple simulations, the clarity of the results can be enhanced and it is possible to gain deeper insights into the underlying dynamics and behavior of the system under investigation. In the previous results it will be useful to clarify the thermal fluctuations effect in the autocorrelation represented in Fig. III.1 and in Fig. IV.2.

VI COMPUTATIONAL DETAILS

The molecular dynamics simulations will be performed using the simulation package NAMD, designed to run simulations of biomolecules in aqueous systems. To run a MD simulation with NAMD, the following files will be given as input. A coordinate file (.pdb) containing initial atomic positions and velocities, in our case, our system consists of 128 DMPC phospholipids and 4377 water molecules. A structure file (.psf) providing structural information and bonding interactions, a force field parameter file (.prm) defining the atom potential, and a configuration file specifying simulation settings and options, in our case we will use CHARMM36 force field, which is parametrized to describe optimally lipids in aqueous solution.

The simulation will be done using Langevin dynamics. Langevin dynamics is a stochastic process that simulates the behavior of particles in a solvent by incorporating random forces and accounting for thermal fluctuations. It combines deterministic and stochastic components to accurately capture the inherent randomness and thermal effects in molecular systems. The stochastic nature of Langevin dynamics reflects the real-world behavior of particles and their interactions with the surrounding environment. For instance, a Langevin thermostat recreates the thermal fluctuations proper of the canonical ensemble, that can not be obtained with other thermostat like the Berendsen based on rescaling velocities. Moreover, the bonds will take as rigid, NAMD uses RATTLE algorithm to do so, that is an method based on SHAKE algorithm with some modifications. To thoroughly investigate the thermodynamic states of a system, we follow a two-step simulation

approach. Each state is explored through the following simulations:

1. **Equilibration Simulation:** We begin by conducting an equilibration simulation run. This simulation is designed to bring the system from its initial conditions to the desired thermodynamic state. In our case, we opt for an NPT equilibration run, where we aim to establish an equilibrium state with specific temperature and pressure. This equilibration run allows the system to reach a stable configuration. The Langevin thermostat and barostat are used.

The equilibration simulation helps to ensure that the system is in a reliable and representative state before proceeding with the analysis. It allows for the relaxation of any initial biases or imbalances and ensures that the system has reached a state consistent with the desired temperature and pressure. The isobaric isothermic ensemble is chosen for the equilibration because it is close to experimental set up. This is done in simulation by including a thermostat and a barostat.

2. **Production Simulation:** Following the equilibration simulation, we move on to the production simulation run. This simulation departs from the final state achieved during the equilibration phase. Its primary purpose is to generate the necessary data for further analysis and to obtain meaningful results.

To simplify the analysis of the trajectories, we choose to perform a production run with a fixed simulation box size and temperature, known as NVT (canonical ensemble). By keeping the box size fixed, we eliminate the influence of volume fluctuations on the results. That implies that the membrane will not change its size during the simulation. Additionally, maintaining a constant temperature throughout the production run allows to sample the phase diagram in a isothermal.

During the production simulation run, we collect data on various properties, such as particle positions, velocities, and energies. This data forms the basis for analyzing the system’s behavior, studying its thermodynamic properties, and extracting relevant information or observables.

By conducting both an equilibration simulation and a subsequent production simulation, we ensure that the system has reached a state representative of the desired thermodynamic conditions. The equilibration phase allows for the system to relax and adjust, while the production phase generates data for analysis.

Both phases of the simulation will be done with periodic boundary conditions. About the parameters used in the simulation, the equilibration will use a time step of 2 fs and do 1000000 iterations, that is 2 ns. Furthermore,

the barostat will be settle to $P = 1\text{atm}$. The temperature of the thermostat will be the one corresponding to the desired temperature of the simulation. In the production phase, it uses a time step of 2 fs and do 5000 iterations, that is 10 ps. In this part the barostat is not introduce but there is a thermostat.

VII CONCLUSIONS

In conclusion, our study focused on a lipid bilayer system with water molecules solvating the membrane and forming hydrogen bonds with the hydrophilic heads. We investigated the effect of temperature variation taht generates thermal fluctuations noticeable in the dipole rotational autocorrelation of water molecules. Our analysis considered different distances range from the membrane to explore the interaction between water and lipid heads.

We found that water molecules in closer proximity to the lipid bilayer exhibited higher dipole rotational autocorrelation. This observation suggests a short-range interaction between water and the lipid heads. Additionally, as temperature increased, we observed a decrease in dipole rotational autocorrelation across all distance ranges. Higher temperatures resulted in reduced correlation, indicating faster relaxation of water dipoles.

Furthermore, by performing multiple simulations with different initial conditions under the same thermodynamic settings, we demonstrated the importance of repetition in smoothing the autocorrelation curve. This repetition helped to average out stochastic fluctuations and provided a more robust representation of the system's dynamic be-

havior, consistent with the nature of stochastic processes.

In summary, our findings highlight the influence of temperature on dipole rotational autocorrelation in a lipid bilayer system. Closer proximity to the membrane enhances the autocorrelation, while higher temperatures decrease it. The repetition of simulations with different initial conditions confirms the smoothing effect and underscores the significance of stochastic processes in understanding the dynamics of the system. Overall, these insights contribute to a deeper understanding of the behavior and interactions within lipid bilayers in relation to water molecules.

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